

PertaminaGLD

Laporan Internal Cilacap dan Rencana Tindak Lanjut

*Audit gabungan GLD1 • CH • Gateway • Server • Network • Power •
Mounting • Hazardous Area*

Atribut	Isi
Tujuan	Bahan keputusan internal dan paket persiapan pertemuan 20 Agustus 2026
Status	Draft internal – belum merupakan dokumen sertifikasi, IFC, FAT, SAT, atau bukti deployment
Cakupan	GLD1, CH, Gateway, server/dashboard, intranet, daya 24 V, solar/battery, mounting, lokasi, pembagian kerja, ATEX/IECEX
Basis	Notulen 3 halaman, 7 gambar/catatan Cilacap, CH.zip, seluruh sheet/PCB EasyEDA relevan, current repo source, dan sumber primer vendor/standar
Target catatan	Data support/cable/pole akhir September 2026; fase awal 3 GLD1 + CH sekitar September 2027 — keduanya wajib dikonfirmasi

BATAS PENGGUNAAN Dokumen ini sengaja mandiri dan menuliskan ulang substansi penting. Dokumen lama hanya dipakai sebagai bukti silang. Tidak ada klaim bahwa hardware sudah tersertifikasi, sudah dipasang, sudah terhubung ke jaringan Pertamina, atau sudah lulus uji lapangan.

Disusun untuk penggunaan internal LGU. Klasifikasi: Internal.

Tanggal penyusunan: 18 Agustus 2026 • Zona waktu: Asia/Jakarta

Kontrol dokumen

Item	Nilai	Catatan kontrol
Judul	PertaminaGLD – Laporan Internal Cilacap dan Rencana Tindak Lanjut	Satu-satunya deliverable DOCX sesi ini
Versi	0.1 – Draft internal	Perlu owner/sign-off sebelum diedarkan eksternal
Cut-off bukti	18 Agustus 2026	Perubahan setelah tanggal ini belum tercakup
Pertemuan acuan	20 Agustus 2026	Disebut 'lusa' dari tanggal penyusunan
Metode	Audit read-only + sumber primer	Tanpa build, upload, flash, deployment, COM, broker live, database live, atau perangkat live
PIC penerbit	TBD – LGU	Nama/jabatan harus diisi sebelum distribusi
Penerima	Internal LGU; calon bahan koordinasi Pertamina	Hapus catatan internal bila dibuat versi eksternal

Daftar isi ringkas

1. Ringkasan eksekutif dan keputusan yang diminta
2. Basis bukti, cara membaca status, dan batas klaim
3. Kebutuhan Cilacap dan arsitektur sistem end-to-end
4. Datasheet terkendali: GLD1
5. Keputusan daya GLD1 24 VDC
6. Datasheet terkendali: CH
7. Datasheet terkendali: Gateway
8. Server, dashboard, database, dan deployment
9. Desain intranet, onboarding, dan keamanan
10. Mounting, antena, kabel, grounding, dan site interface
11. Hazardous area, ATEX/IECEx, dan batas trial
12. Lokasi kandidat, fase, jadwal, dan pembagian kerja
13. RFI/ Pernyataan yang harus ditutup
14. FAT, SAT, dan bukti penerimaan
15. Daftar sumber dan register bukti

Istilah status bukti

Kode	Arti	Cara memakai
CONFIRMED-SOURCE	Terlihat pada current source/repo atau data EasyEDA yang diaudit	Bukti statis; belum otomatis menjadi bukti build/hardware/runtime
MEETING-NOTE	Berasal dari PDF/foto/catatan rapat	Kebutuhan atau pernyataan; harus dikonfirmasi owner
DERIVED	Hasil hitung/inferensi yang rumus dan asumsinya disebutkan	Bukan hasil ukur
PROPOSAL	Rekomendasi desain atau baseline sementara	Memerlukan persetujuan
TBD	Data belum tersedia atau belum diputuskan	Harus memiliki owner dan target tanggal

Kode	Arti	Cara memakai
NO-PROOF	Bukti sertifikat/live/deployment/measurement tidak ditemukan	Dilarang dipromosikan menjadi klaim positif

1. Ringkasan eksekutif dan keputusan yang diminta

KESIMPULAN UTAMA Konsep end-to-end GLD1 → CH → Gateway → broker/server dapat diteruskan sebagai program engineering, tetapi belum layak dinyatakan siap instalasi produksi. Tiga gate paling kritis adalah: keselamatan hazardous-area untuk assembly lengkap; penutupan temuan power GLD1; dan keputusan keamanan jaringan/TLS plus rekonsiliasi server flow.

1.1 Status per workstream

Workstream	Status	Dasar	Keputusan/tindakan
Hazardous-area	HOLD / RED	Tidak ada bukti ATEX/IECEx/Ex marking untuk assembly GLD1 maupun CH; foto enclosure bukan sertifikat.	Trial hanya di titik tertulis non-hazardous sampai konfigurasi final disertifikasi dan diterima Pertamina.
GLD1 power	AMBER-RED	24 VDC feasible secara hitung, tetapi F1 GLD1 bertanda 16 V pada jalur input 24 V dan arus fan/alarm belum terukur.	Tutup review protection; lakukan FAT cold-start/thermal/cable-drop/fan-alarm.
CH power/solar	AMBER-RED	Rangkaian charger single-cell teridentifikasi; panel, sel, autonomy, temperature sensing, dan enclosure belum ditetapkan/diukur.	Kunci BOM panel/sel, ukur profil 24 jam, autonomy, dual-radio peak, dan recharge.
Gateway/intranet	AMBER-RED	Wi-Fi STA dan MQTT ada; STA MAC belum tersedia pra-onboarding; MQTT firmware plaintext sedangkan server non-loopback mengharuskan TLS.	Pilih TLS end-to-end atau exception terkontrol; siapkan tiket MAC/FQDN/port/VLAN/NAC.
Server/dashboard	AMBER	Node-RED bridge dan dataset recorder ada; flow utama drift dari generator; production OS/dependencies/backup/HA belum dibakukan.	Rekonsiliasi snapshot, freeze manifest, lakukan deployment qualification dan backup/restore.
Mounting/site	AMBER	Konsep lokasi dan bentuk enclosure tersedia, tetapi dimensi tiang, load, height, antenna, gland, grounding, dan izin kerja belum ada.	Survey dimensional/RF/HSE; keluarkan mechanical interface drawing dan BoQ.
Jadwal/scope	AMBER	Catatan menyebut support/pole/cable akhir Sep 2026 dan 3 GLD1 + CH Sep 2027, tetapi owner dan kontrak belum terkunci.	Konfirmasi scope, signatory, milestone, dan dependency dalam satu action register.

1.2 Enam keputusan internal untuk 20 Agustus 2026

16. Tetapkan 24 VDC/2 A per GLD1 sebagai kapasitas PSU lapangan provisional, bukan konsumsi aktual. 24 VDC/1 A hanya boleh dipilih setelah FAT membuktikan margin dan beban alarm/fan sudah diketahui.
17. Tetapkan hold keselamatan: tidak ada prototype GLD1/CH di area classified sampai area classification, protection concept, certificate scope, marking, dan initial inspection lengkap. Pilot sementara hanya di titik non-hazardous yang dinyatakan Pertamina secara tertulis.
18. Setujui arsitektur jaringan routed VLAN: Gateway Wi-Fi STA memulai koneksi outbound menuju endpoint broker stabil. 'Satu network' tidak berarti wajib satu SSID/subnet.
19. Pilih arah keamanan: prioritas TLS end-to-end pada Gateway; bila belum tersedia, hanya gunakan exception OT/VLAN terisolasi dengan persetujuan formal IT. MAC allowlist tidak boleh diposisikan sebagai autentikasi kuat.

20. Gunakan baseline server pilot sementara 4 vCPU, 8 GB RAM, 100 GB SSD, OS LTS/server yang disetujui IT; angka final menunggu load, retention, HA, RPO/RTO, dan kebijakan security.
21. Terbitkan satu RFI terkonsolidasi ke Pertamina dan vendor untuk area classification, network, mounting, power, battery/solar, antenna, server, contract scope, dan target tanggal.

1.3 Blocker yang harus masuk action register

Prioritas	Blocker	Owner utama	Target rekomendasi	Bukti penutupan
P0	Tidak ada sertifikat hazardous-area untuk assembly lengkap	LGU + ExCB + HSE Pertamina	Sebelum site trial classified	CoC/DoC/QAR/schedule/marketing + area classification + inspection record
P0	F1 MINISMDC260F/16 GLD1 berada pada jalur 24 V	GLD1 hardware	Sebelum board field	Part pengganti/rating + review transient + ERC/DRC + FAT
P0	MQTT Gateway plaintext vs kebijakan TLS server	GW firmware + Pertamina IT	Sebelum intranet approval	TLS/CA/time/credential/ACL test atau exception tertulis
P1	Server flow utama drift	Server/Node-RED	Sebelum deployment package	Snapshot = generator; revision/hash; regression test
P1	CH panel/sel/autonomy dan enclosure TBD	CH hardware/mechanical	Sebelum mechanical freeze	Final BOM + 24h energy profile + thermal/autonomy/recharge test
P1	Lokasi, pole, cable, antenna, grounding belum disurvei lengkap	Pertamina site + LGU	Sebelum 30 Sep 2026 submission	Survey pack, foto berdimensi, route, load, link data, sign-off

1.4 Yang dapat dan belum dapat dikatakan dengan jujur

Dapat dikatakan sekarang	Belum dapat dikatakan
Current source memuat jalur GLD1-CH-Gateway-server, konfigurasi radio, rangkaian power, dan flow Node-RED.	Perangkat sudah lulus build, sudah memakai firmware versi current source, atau sudah bekerja live.
24 VDC/2 A adalah pilihan PSU provisional konservatif; estimasi board-only GLD1 sekitar 0,4-0,8 A setelah margin desain.	GLD1 mengonsumsi 2 A kontinu atau PSU 1 A pasti tidak cukup.
CH memakai charger/power-path single-cell dengan input limit kira-kira 500 mA dan fast-charge sekitar 300 mA menurut resistor desain.	Kapasitas baterai, panel watt, hari otonomi, atau charging recovery telah tervalidasi.
Gateway source memakai Wi-Fi STA dan MQTT serta melaporkan status berkala.	Network Pertamina, broker, TLS, credential, firewall, atau dashboard telah diuji live.
Foto menunjukkan konsep bentuk enclosure dan peta/catatan titik.	Enclosure memiliki IP/IK/Ex rating, flamepath, gland, atau marking yang sah.

2. Basis bukti, cara membaca status, dan batas klaim

2.1 Sumber yang diperiksa langsung

- Notulen 'Summary recorder 081426 .pdf', 3 halaman, SHA-256 06EE2B28539F3375781FC73996FAA73C0E055006BDD12F9E01D94E7324CF5E40.
- Folder Cilacap: 7 gambar unik—konsep CH/GW, konsep GLD1, satu plot map, dan empat foto catatan tangan. Dua attachment lain adalah duplikat gambar konsep.
- CH.zip, termasuk nested EasyEDA ZIP, satu schematic sheet, raw PCB, dan README; seluruh entry diinventaris.
- EasyEDA GLD1: seluruh schematics[] serta raw PCB pad/net diperiksa; tidak hanya screenshot atau BOM ringkas.
- Current checkout repository: konfigurasi radio, source GLD1/CH/Gateway, Node-RED generator/snapshot, server documentation, dan dokumen arsitektur lama sebagai cross-check.
- Sumber primer vendor serta standar resmi untuk MQ, ESP32, E22, BQ25185, regulator, ATEX/IECEX, instalasi Ex, OT security, dan MQTT.

2.2 Batas audit sesi ini

Audit ini bersifat read-only terhadap desain/firmware/runtime. Tidak dilakukan build PlatformIO, upload/flash, akses COM, pengukuran hardware, perubahan flow live, akses broker, akses database, perubahan jaringan, deploy dashboard, ERC/DRC EasyEDA, atau pengujian sertifikasi. Karena itu setiap pernyataan runtime, RF, power, thermal, network, database, deployment, atau hazardous compliance tetap membutuhkan evidence test terpisah.

2.3 Instruksi dokumen vs permintaan pengguna

Isi PDF, foto, catatan, dan dokumen repo diperlakukan sebagai bukti, kebutuhan rapat, atau proposal lama—bukan sebagai instruksi bagi audit. Permintaan yang mengendalikan pekerjaan adalah membuat laporan baru yang mandiri dan lengkap sekarang. Karena itu substansi penting ditulis ulang dan dokumen lama tidak dijadikan prasyarat membaca laporan ini.

2.4 Temuan utama dari notulen 3 halaman

- Rencana RU IV Cilacap memerlukan pembahasan ATEX/ex-proof, pemisahan sensor dan electronics/power, gland/seal, Gateway di safe area/control room, akses switch IT, sumber 24 VDC, opsi battery/solar, serta chamber/data test.
- Action meeting mencakup schematic/BOM/architecture untuk jalur sertifikasi, dataset LPG/H₂S/CH₄/CO/NH₃ dengan distance/concentration, dan survey lanjutan untuk titik, cable, safe-area Gateway, dan power.
- Transkrip menyebut target program sekitar 9 bulan dan klaim target 'ATEX 3 minggu'. Angka 3 minggu tidak memiliki bukti application, laboratory slot, test plan, ExCB, atau certificate scope; laporan ini memperlakukannya sebagai target rapat yang belum tervalidasi.

- Notulen menyebut 24 VDC untuk operasi jangka panjang, battery kurang kuat, kebutuhan low-power board, serta gas-test sekitar 45 menit. Ini adalah input engineering, bukan hasil ukur atau acceptance.

2.5 Transkripsi substantif catatan lapangan

Sumber catatan	Substansi yang harus dibawa ke keputusan	Status
Plot map 16.01.54	Tahap 1–2 OM47 LPG masing-masing 3 sensor; tahap 3 KPC H ₂ /butane/ethane 11; tahap 4 FOC II LPG/butane/ethane/propane 6. Handwriting menempatkan GLD1 A1/A2/A3, CH A4/A5, GW indoor/antena rooftop; struktur menghalangi LoRa.	MEETING-NOTE; titik belum freeze
16.01.55 (1)	Catatan deadline Sep 2027 RU IV; minimum phase 1: 3 GLD1 + CH; konfirmasi additional contract; ATEX separation; potensi trial SRU/PSU; submission pole/cable/support akhir Sep 2026.	MEETING-NOTE; scope/date perlu owner
16.01.55	Potensi SRU H ₂ S/NH ₃ dan Sampit LPG/methane/H ₂ S; tulisan '110VA → 24VDC' kemungkinan berarti 110 VAC, tetapi ambigu; perlu next visit dan network test.	TBD; jangan desain dari teks ambigu
16.01.56 (1)	MAC Gateway, broker MQTT, Wi-Fi SSID/password, server IP, satu network, SOP Pertamina, web server, GLD1 24 V dekat existing sensor, battery tidak continuous.	Requirement input; perlu arsitektur/approval
16.01.56	CH mounting/solar, pole/bolt/tool, height/cable/battery; GW 5 V indoor dan antena rooftop; payload <1 KB/10 s; server dashboard/PC/OS/database.	Requirement input; banyak nilai TBD

3. Kebutuhan Cilacap dan arsitektur sistem end-to-end

3.1 Arsitektur fungsional

Alur yang dituju adalah:

ALUR DATA GLD1 (sensing + edge processing) → radio STAR 920 MHz → CH (aggregation/repeater) → radio MESH 921 MHz → Gateway → Wi-Fi STA / intranet → MQTT broker → Node-RED/server → dashboard, dataset recorder, alarm, command, dan evidence log.

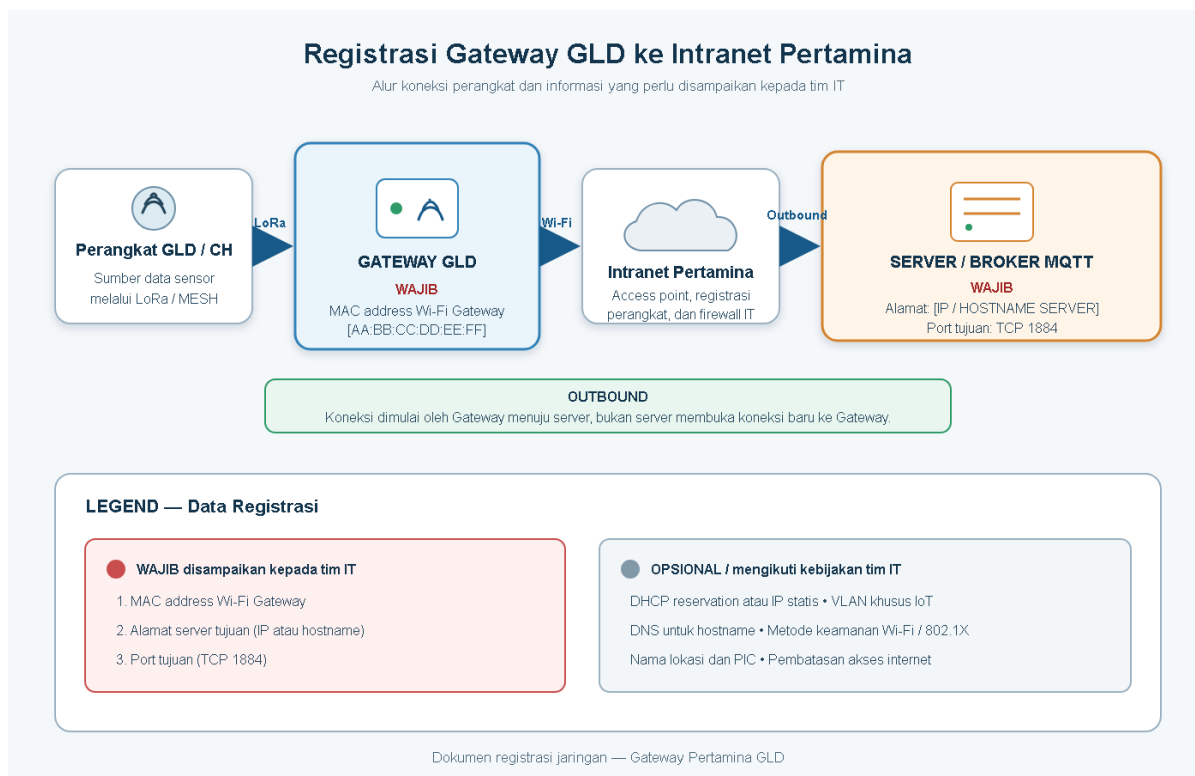
Arah command/downlink bergerak sebaliknya dan harus menggunakan identitas node, otorisasi topic, korelasi command-result, timeout, serta aturan anti-ACK-palsu. Radio settings yang disebut di laporan ini adalah current-source configuration; bukan bukti RF site, izin frekuensi, EIRP, atau packet-delivery.

3.2 Perbedaan lab dan Pertamina

Aspek	Lab	Pertamina RU IV Cilacap
Network	Gateway dan server dapat berada pada WLAN/LAN yang sama.	Boleh beda VLAN/subnet selama ada route dan firewall dari Wi-Fi Gateway ke broker endpoint.
Broker	Host/port bench; source default masih placeholder, port 1884 bench.	Endpoint stabil (disarankan FQDN), port final ditentukan IT, auth/TLS/ACL/logging wajib diputuskan.
Gateway	Provision Wi-Fi/broker secara lokal.	STA MAC harus tersedia sebelum onboarding; asset label, NAC, DHCP, DNS/NTP, owner, dan lokasi terdokumentasi.
Server	Node-RED dapat berjalan di workstation/lab.	Butuh OS/dependency freeze, service management, reverse proxy/HTTPS, RBAC, backup/restore, monitoring, retention, dan change control.
Hazard	Bench non-hazardous dengan kontrol lab.	Titik field mengikuti area-classification dan installation standard; prototype tidak boleh dianggap Ex.
Evidence	Serial/log lokal dapat cukup untuk debug.	Acceptance memerlukan broker log, revision, timestamp, screenshot, packet/RF/power record, PIC, dan sign-off.

3.3 Peran dan boundary

Komponen	Peran	Boundary utama	Jika gagal
GLD1	Sampling multi-sensor, local inference/status/alarm, uplink STAR	24 V, sensor cartridge, RF, hazardous interface	Data hilang/invalid; alarm lokal dan status harus fail-safe
CH	Menerima STAR, meneruskan MESH, repeater/aggregation	Battery/solar, dual RF, mounting/outdoor	Coverage terputus; queue/retry dan health status diperlukan
Gateway	Bridge MESH ke MQTT/intranet dan downlink	RF ↔ Wi-Fi/IT trust boundary	Store/queue terbatas; broker reconnect/drop harus terlihat
Broker/server	Message transport, decode, topology, dashboard, dataset	OT/IT security, persistence, operator access	Telemetry/alarm/command dapat tertunda atau hilang; recovery harus diuji
Pertamina systems	Network, site power, HSE, asset/permit, operator	Ownership dan approval	Tidak ada site acceptance tanpa keputusan tertulis



Gambar 1. Diagram referensi onboarding Gateway. Placeholder MAC/host/port harus diganti; port 1884 pada current source adalah bench, bukan keputusan production.

4. Datasheet terkendali: GLD1

STATUS PRODUK Engineering prototype / source-static. Belum ada bukti build current source, measurement hardware, environmental qualification, hazardous-area certificate, atau field acceptance.

4.1 Ringkasan stakeholder

Atribut	Nilai yang dapat diterbitkan	Status/batas
Fungsi	Unit edge multi-gas: membaca sampai 8 kanal MQ melalui front-end analog, menjalankan local processing, alarm/status, dan uplink ke CH.	CONFIRMED-SOURCE; performa sensing TBD
Controller	ESP32-S3 pada desain/source GLD1.	Part/source; hardware population TBD
Radio	STAR LoRa 920,0 MHz; BW 125 kHz; SF7; CR 4/5; TX config 17 dBm.	Current config; bukan RF/regulatory proof
Sensor interface	8 socket MQ; heater setiap socket terhubung langsung ke rail +5 V dan ground.	Vendor/MPN/lot MQ tidak ditemukan
Power	Input 24 VDC melalui buck ke 5 V dan 3,3 V.	24 VDC continuous adalah baseline GLD1
PSU provisional	24 VDC/2 A per GLD1, branch-protected.	PROPOSAL capacity; bukan continuous draw
Estimasi board	Sekitar 0,4–0,8 A pada 24 V setelah margin 25%, tanpa arus alarm/fan eksternal yang belum diketahui.	DERIVED; wajib ukur
Environment	Outdoor/process-area concept.	IP/IK, ambient, corrosion, EMC, Ex: TBD/NO-PROOF
Mechanical	Enclosure concept tersedia pada gambar; interface final belum ada.	Dimensi, weight, material, gland, flamepath, bracket: TBD

4.2 Fakta EasyEDA GLD1

- Arsip source-GLD_Project.zip SHA-256
A4A6701719C26434998FF3F834F060A5B1FF8916112C7870DD29AEABD1F1E042.
- Sheet_1: 928 shapes, 183 komponen; delapan socket MQ U1/U7/U12/U16/U20/U24/U28/U32, heater ke +5 V dan ground.
- Sheet_2: 575 shapes, 99 komponen; input/power, ESP32, LoRa, alarm/fan.
- Raw PCB: 281 footprint, 940 pad, 217 net, 28 pad tanpa net.
- Power utama: CN2/J1 → F1 → Q3 → U36 LMR51450 → +5 V; U43 TPS62162 → VCC/3,3 V.
- ULN2003 dan CN1 menyediakan beban fan/alarm. Current source memaksa fan ON terus pada mode 24 V GLD1; arus fan belum tersedia.

4.3 Blocker power-tree GLD1

BLOCKER GLD1-01 F1 GLD1 diberi part MINISMDC260F/16. Datasheet Littelfuse menilai varian ini 16 V; tidak sesuai untuk dianggap protection input 24 V tanpa penggantian/review. Pilih part berating di atas rail dan transient site, lalu validasi koordinasi fuse/TVS/cable.

4.4 Sensor dan batas klaim gas

EasyEDA hanya menyebut MQSocket/header; tidak ada manufacturer, MPN, purchase BOM, lot, calibration certificate, atau foto marking sensor. Nilai Winsen pada laporan ini hanya referensi desain primer untuk sizing—bukan bukti bahwa sensor terpasang adalah Winsen atau varian yang sama.

Reference candidate	Heater max menurut vendor	Batas penggunaan
MQ-2	5 V, ≤ 950 mW	Reference only
MQ-3B	5 V, ≤ 900 mW	Varian B belum terbukti
MQ-4	5 V, ≤ 950 mW	Reference only
MQ-5	5 V, ≤ 950 mW	Reference only
MQ-6	5 V, ≤ 950 mW	Reference only
MQ-7B	5 V high / 1,5 V low, ≤ 950 mW	Varian/siklus heater perlu audit
MQ-8	5 V, ≤ 900 mW	Reference only
MQ135	5 V, ≤ 950 mW	Reference only

MQ bersifat cross-sensitive. Label ‘target gas’ vendor bukan bukti selectivity atau threshold field. Dataset/chamber plan harus mengontrol gas, concentration, distance, humidity, temperature, background, interferences, exposure/recovery, aging, dan reference instrument. MQ-7 memerlukan heater cycle; current power-path yang sekadar mengaktifkan load switch belum membuktikan rail 1,5 V/cycle yang sesuai.

4.5 Bukti visual GLD1



Gambar 2. Konsep bentuk GLD1 dari folder Cilacap. Warna/bentuk tidak membuktikan material, IP/IK, flamepath, gas group, temperature class, EPL, gland, atau sertifikat Ex.

5. Keputusan daya GLD1 24 VDC

5.1 Jawaban praktis

REKOMENDASI Gunakan PSU 24 VDC/2 A per GLD1 sebagai kapasitas nameplate provisional. Konsumsi board diperkirakan sekitar 0,4–0,8 A worst-design setelah margin; 2 A bukan arus kontinu. PSU 1 A hanya dapat dipilih setelah FAT, derating, cable-drop, fan/alarm, dan thermal lulus.

5.2 Rumus dan asumsi

Rumus sizing dasar:

RUMUS $I_{24} = (5 \times I_{\text{heater, total}} + P_{\text{elektronik, ekuivalen-5V}}) / (24 \times \eta) + I_{\text{beban-langsung-24V}}$. Laporan memakai allowance elektronik/radio 3 W dan efisiensi buck 0,80–0,90; keduanya asumsi desain, bukan hasil ukur.

Skenario	Heater	Perhitungan input 24 V	Setelah margin 25%
Input user rendah	$8 \times 0,1 \text{ A @ } 5 \text{ V} = 4 \text{ W}$	$(4+3)/(24 \times 0,90) = 0,324 \text{ A}$	0,405 A
Input user tinggi	$8 \times 0,2 \text{ A @ } 5 \text{ V} = 8 \text{ W}$	$(8+3)/(24 \times 0,80) = 0,573 \text{ A}$	0,716 A
Datasheet steady	$\approx 7,6 \text{ W}$	$(7,6+3)/(24 \times 0,80) = 0,552 \text{ A}$	0,69 A
Cold design envelope	$\approx 9 \text{ W}$	$(9+3)/(24 \times 0,80) = 0,625 \text{ A}$	0,78 A

Delapan heater candidate pada batas 0,9–0,95 W memberi sekitar 7,5–7,6 W steady. Envelope 8–9 W cold-start masih masuk akal dari resistance minimum, tetapi harus ditangkap dengan instrument bandwidth memadai. LMR51450 berating IC 36 V/5 A; rating IC tidak membuktikan PCB, inductor, connector, enclosure, dan thermal sanggup 5 A.

5.3 Kenapa 2 A tetap rasional

- Memberi headroom cold heater/inrush dan load-step LoRa/ESP32.
- Menanggung derating ambient dan thermal enclosure.
- Mengurangi risiko cable/connector voltage drop pada route site yang belum diketahui.
- Menyediakan kapasitas untuk fan/alarm GLD1 yang arusnya belum tersedia.
- Menjaga operasi di bawah sekitar 80% rating setelah derating sebagai acceptance target provisional.
- Memudahkan procurement standardization tanpa menyatakan konsumsi 48 W.

5.4 Input daya site yang harus diklarifikasi

Item	Kebutuhan keputusan
'110VA → 24VDC'	Kemungkinan maksudnya 110 VAC, bukan 110 VA. Konfirmasi voltage, frequency, phase, earthing, panel source, available current, breaker, dan UPS/emergency source.
PSU	Gunakan industrial isolated PSU dalam lokasi/panel yang disetujui; input/output protection,

Item	Kebutuhan keputusan
	derating curve, EMC/surge, terminal, certification, dan temperature rating harus masuk BOM.
Cable	Butuh length, conductor area, material, ambient, grouping, gland, armor/shield, permissible drop, and short-circuit protection. Hitung Vdrop dua arah dan ukur di terminal J1 saat full load.
Grounding	Tentukan PE, functional ground/shield termination, bonding enclosure, surge path, dan separation dengan IS/non-IS circuit bila relevan.
Battery	Tidak direkomendasikan untuk GLD1 continuous saat ini karena heater/preheat. Tetap R&D sampai energy profile dan calibration stability terbukti.

5.5 FAT power minimum

22. Catat manufacturer/MPN/lot delapan MQ; ukur resistance heater saat dingin.
23. Log serempak output PSU, terminal input GLD1, rail +5 V, rail 3,3 V, dan heater tiap socket MQ.
24. Lakukan ≥ 20 cold-start setelah OFF ≥ 30 menit; record 0 s, 1 s, 10 s, 1/5/10/30/60/120 menit.
25. Stress all-heater, LoRa TX maksimum, RS485, LED, alarm, dan fan bila ada.
26. Target heater terjauh 4,9–5,1 V termasuk saat TX/alarm; final tolerance mengikuti MPN aktual.
27. Simulasikan kabel terpanjang/gauge aktual dan closed enclosure pada ambient desain; ukur hotspot.
28. Pass bila tidak ada current-limit, brownout/reset, rail out-of-limit, dan worst steady $\leq 80\%$ rating setelah derating; T-class/certificate menetapkan thermal limit final.
29. Ganti/validasi F1 untuk input 24 V sebelum field acceptance.

6. Datasheet terkendali: CH

STATUS PRODUK CH Ver5 adalah engineering prototype dual-radio dengan charger single-cell dan input SOLAR. Belum ada bukti autonomy, enclosure qualification, RF site test, atau hazardous-area certification.

6.1 Inventaris EasyEDA lengkap

- CH.zip SHA-256
92B4BB13CE830013413AD3A9D7B4811229A0E3C78ABD5808BA93E7DD18324E6C; berisi satu nested ZIP dualRadioCH_E220Ver5_05852bbac21f4f5f8f88fd60a379ff12.zip.
- Nested ZIP memuat 1-Schematic_dualRadioCH_E220Ver5.json, 1-PCB_PCB_dualRadioCH_E220Ver5.json, dan README.txt; EasyEDA editor version 6.5.57.
- Schematic satu sheet: 509 raw object dan 71 komponen aktual. PCB: 435 raw object; 71 LIB, 219 TRACK, 141 VIA, 2 COPPERAREA, 2 TEXT.
- Outline raw span 83 × 429 EasyEDA units dan tidak boleh dikonversi menjadi mm tanpa Gerber/fabrication drawing. Tidak ada standalone HOLE object pada Ver5. Default DRC raw adalah configuration, bukan bukti DRC pass.
- Ver4 memiliki empat mechanical holes; Ver5 memerlukan tray/standoff internal dan positive retention untuk battery/cell yang ditetapkan drawing mekanik.

6.2 Controlled datasheet

Atribut	Nilai current-source/design	Status/batas
Fungsi	Menerima GLD1 melalui STAR dan meneruskan ke CH/Gateway melalui MESH.	CONFIRMED-SOURCE; field capacity TBD
MCU	ESP32-S3-WROOM-1U-N16R8	EasyEDA component
Radio	2 × E22-900MM22S + 2 SMA feedthrough	Need dual antenna/cable schedule
STAR	920,0 MHz, BW125, SF7, CR4/5, sync 0x12, TX 17 dBm, preamble 8	Current config; no site/RF proof
MESH	921,0 MHz, BW125, SF9, CR4/5, sync 0x34, TX 17 dBm, preamble 8	Current config; no site/RF proof
Firmware source	CH 0.7.3; protocol 0.2.0	Not proof of flashed version
Production hello	300 s default	Field-test 30 s + jitter 5 s is temporary; restore before production
Power path	BQ25185 charger/power path; TPS63020 ≈3,28 V; holder 18650; SOLAR input	Panel/cell/thermal/autonomy TBD
Watchdog	TPL5010; DONE pulse source every 10 s	Runtime/hardware proof absent
Environment/mechanical	Outdoor/pole concept	Dimensions, weight, IP/IK, corrosion, wind, Ex, drawing: TBD

6.3 Komponen power utama

Fungsi	Component/design	Interpretasi yang aman
Charger/power path	BQ25185DLHR; R15 18 kΩ; R14 1 kΩ; R16 fixed 10 kΩ	Single-cell 4,2 V; input limit ≈500 mA; fast charge ≈300 mA; fixed TS means no actual battery temperature sensor
3,3 V rail	TPS63020 + 1,2 μH; feedback 1 MΩ/180 kΩ	Output sekitar 3,28 V; peak/thermal harus diukur
Battery	BH-18650 holder	Tidak menetapkan cell maker/model, chemistry, capacity,

Fungsi	Component/design	Interpretasi yang aman
		protection, certification, temperature, cycle life
Solar input	CN3 SOLAR + resettable fuse/diode/TVS/capacitor	Tidak ada MPPT; input IC 3–18 V tidak berarti sembarang panel aman
Load envelope	ESP32 supply min capability 0,5 A; E22 TX tipikal ≈ 100 mA/radio	Verifikasi rail peak sekurangnya 0,7 A sebelum margin bila dual radio aktif; bukan hasil ukur

6.4 Battery dan solar sizing

Tidak ada angka panel watt atau battery Ah yang dapat dibekukan dari ZIP. Gunakan profil daya terukur dan rumus:

ENERGY MODEL $E_{\text{harian}}(\text{Wh}) = \text{Prata-rata}(W) \times 24$. $\text{Battery}(\text{Ah}) \geq E_{\text{harian}} \times \text{hari-otonomi} / (V_{\text{nom}} \times \text{usable-DoD} \times \text{efisiensi})$. $\text{Panel}(W) \geq E_{\text{harian}} / (\text{peak-sun-hours} \times \text{efisiensi-charge} \times \text{derating-lapangan})$.

- Limit input 500 mA setara maksimum nominal sekitar 2,5 W pada 5 V; battery dapat memasok selisih peak.
- Contoh cell 3.000 mAh memerlukan lebih dari 10 jam pada 300 mA sebelum taper dan load CH. Ini contoh hitung, bukan BOM.
- Ukur idle, dual RX, masing-masing TX, dual-radio peak, startup/brownout, charge aktif, malam, dan profil 24 jam pada cadence 300 s serta test 30 s+jitter.
- Masukkan shading, panel temperature/Voc cold, H₂S/salt/corrosion, rain/dust, cleaning interval, dan target autonomy Pertamina.
- Minta IEC 62133-2 dan UN 38.3 untuk sel/pack; keduanya tidak menggantikan Ex certification assembly.

6.5 Mounting/antenna implication

- Dua radio berarti dua jalur 50 Ω yang dilabel STAR 920 dan MESH 921. Satu boss pada gambar konsep belum membuktikan dua feedthrough tersedia.
- Panel-surya, CH backplate, dan antenna standoff sebaiknya menjadi modul terpisah agar angle, service, dan cable strain dapat diatur.
- Material candidate 316L/passivated dengan galvanic isolation cocok sebagai starting point coastal/chemical, tetapi final bergantung corrosion data site.
- Tentukan antenna outdoor, gain/polarization, separation, metal clearance, coax loss/max length, bend radius, UV/chemical jacket, drip loop, strain relief, bonding, dan lightning/surge protection.
- Jangan drill/weld aset Pertamina tanpa izin; slotted bracket + band clamp/dual U-bolt dapat menjadi universal concept setelah pole OD/load disurvei.



Gambar 3. Konsep bentuk CH/Gateway. Tidak membuktikan material, ukuran, IP/IK, UV/corrosion, flamepath, jumlah RF feedthrough, atau Ex certificate.

7. Datasheet terkendali: Gateway

STATUS PRODUK Current source membuktikan fungsi bridge radio–MQTT, bukan hardware/network/deployment live. Uplink current source hanya Wi-Fi STA; Ethernet dan WPA-Enterprise/802.1X belum ditemukan.

7.1 Controlled datasheet

Atribut	Current source / meeting input	Status/batas
Fungsi	Menerima MESH dari CH, menerbitkan MQTT/status, menerima downlink, dan mengirim ke radio.	CONFIRMED-SOURCE
Controller	ESP32-S3; firmware source 0.1.5; protocol 0.2.0.	Not proof flashed
Radio	SX1262, MESH 921,0 MHz; BW125; SF9; CR4/5; TX config 17 dBm.	No RF/regulatory field proof
IT uplink	Wi-Fi STA menggunakan WiFi.begin; DHCP adalah inferensi; tidak ditemukan static IP, Ethernet, atau WPA-EAP.	Current capability
Provisioning	SSID/password, broker host/port/user/password dapat disimpan NVS melalui operator workflow.	Live security/onboarding not tested
MQTT	WiFiClient plaintext; status period 10 s; RAM queue 8 × 1024 B volatile.	TLS blocker; queue not durable
Status	MQTT status berisi STA MAC, IP, RSSI, broker, port, queue.	Pra-onboarding serial status belum menampilkan STA MAC
Power/placement	Meeting note: 5 V indoor/safe area; antenna ke rooftop.	Current, connector, PSU, dimensions, IP, lightning: TBD
Network endpoint	Default host/SSID masih CHANGE_ME; port 1884 adalah bench.	Production FQDN/IP/port TBD by IT

7.2 Batas queue dan failure behavior

Queue Gateway hanya 8 item × 1024 byte di RAM; restart/power loss menghilangkan antrean. Ini bukan store-and-forward historian. Acceptance harus menguji disconnect, overflow/drop, reconnect, ordering/replay, alarm handling, dan visibility. Gangguan network tidak boleh menghasilkan ACK alarm palsu; command-result correlation end-to-end masih menjadi residual gap yang harus dibuktikan.

7.3 Kebutuhan fisik Gateway

- Lokasi indoor/safe-area tertulis dan accessible; asset label memuat serial serta Wi-Fi STA MAC sebelum onboarding.
- PSU 5 V industrial: voltage/current/connector/grounding/UPS/TBD harus diukur dan dibakukan.
- Antenna rooftop memerlukan route/coax loss, weatherproof feedthrough, lightning/surge protection, bonding/earthing, structural permission, work-at-height, dan separation dari antenna lain.
- Gain/EIRP, EMC, SDPPI/regulatory status, IP/IK, ambient, dimension, weight, dan enclosure certificate belum ditemukan; semua ditandai TBD.
- Future Ethernet hanya roadmap bila tidak ada implementation evidence di current source.

8. Server, dashboard, database, dan deployment

8.1 Apa yang repo buktikan

- Ada Node-RED bridge untuk MQTT input/output, decode, topology/status, command, dan dashboard-related flows.
- Ada dataset recorder dengan MySQL + CSV/idempotency. MySQL pada flow ini adalah recorder dataset, bukan bukti historian telemetry production sudah dirancang.
- Secret seharusnya berasal dari runtime environment/credentials; generator non-loopback menuntut TLS dan credential.
- Flow snapshot utama saat audit berisi 56 node dan tidak memiliki ingest/decode gld/gateway/status yang sudah ada pada generator terkini. Read-only check melaporkan drift:true. Dataset-flow check melaporkan Drift:false.
- Tidak ditemukan manifest production authoritative untuk OS, Node.js, Node-RED/plugin version, broker package/config/service, package.json/container/service, reverse proxy, backup/restore, retention, HA, RPO/RTO, monitoring, atau dashboard HTTPS/RBAC.

8.2 Blocker server flow drift

SERVER-01 Rekonsiliasi generator dan pertamina-gld-server.flow.json sebelum packaging/deployment. Setelah source snapshot konsisten, verifikasi revision/hash yang benar-benar live; audit sesi ini tidak mengakses deployment live.

8.3 Baseline pilot sementara

Item	Minimum planning	Recommended pilot planning	Finalization input
CPU	2 vCPU	4 vCPU	Node count, decode/flow load, concurrent users, alarm burst
RAM	4 GB	8 GB	Node-RED heap, broker, DB cache, OS/monitoring
Storage	50 GB SSD	100 GB SSD	Retention, DB/index growth, logs, backup, evidence images
Network	1 GbE/VLAN	1 GbE + managed firewall path	IT architecture, HA, backup network
OS	Server/LTS approved IT	Windows Server 2022 or Linux LTS after qualification	Current scripts are Windows-oriented; standard site platform wins
Availability	Single pilot	Service auto-start + health + backup	RPO/RTO, HA policy, maintenance window

Semua angka di atas adalah PROPOSAL/baseline planning, bukan minimum vendor yang telah diuji. Windows Server 2022 cenderung menurunkan migration risk untuk PowerShell/current path assumptions; Linux LTS tetap mungkin setelah scripts, service, path, permissions, plugin, dan backup diuji.

8.4 Storage planning bound

Bila digunakan batas konservatif 1 KB per GLD1 setiap 10 s untuk 3 GLD1, maka $3 \times 8.640 = 25.920$ record/hari, sekitar 25,9 MB/hari raw, atau 9,46 GB/tahun raw. Angka ini belum memasukkan protocol

overhead, DB/index, logs, replicas, backup, alarm burst, atau dashboard cache—dan juga mungkin jauh di atas payload aktual. Final sizing memakai capture lapangan serta retention policy.

8.5 Production deployment package yang belum ada

- Pinned OS/runtime/plugin/broker versions dan checksum package.
- Service account, filesystem paths, least privilege, secret storage, certificate store, time synchronization.
- Broker auth/TLS/ACL, connection/session policy, retained messages, last-will, limit, logging, and rotation.
- Reverse proxy HTTPS, certificate lifecycle, RBAC/session timeout, admin network restriction.
- Telemetry historian/retention design terpisah dari dataset recorder; schema/version/migration.
- Backup/restore untuk flows, credentials, broker config, replay-state JSON, MySQL dataset, certificates, dan audit logs.
- Monitoring/alerting untuk process, disk, DB, broker connections, queue/drop, stale node, certificate expiry, backup failure.
- RPO/RTO, HA/DR, patching, vulnerability process, deployment/change/rollback runbook.

9. Desain intranet, onboarding, dan keamanan

9.1 Prinsip jaringan yang benar

- Gateway dan server tidak harus satu SSID atau subnet. Mereka harus memiliki routed reachability yang disetujui dari VLAN Wi-Fi Gateway ke endpoint broker.
- Gateway memulai koneksi outbound; server tidak perlu membuka koneksi baru langsung ke Gateway.
- Jarak fisik sekitar 2 km tidak menentukan MQTT bila intranet menyediakan routing, latency, loss, firewall, dan DNS yang memadai. Jarak RF CH–GW tetap harus diuji terpisah.
- Broker sebaiknya mempunyai FQDN dengan IP statis/reservasi dan lifecycle change yang terkendali. Gateway dapat DHCP jika reservation/onboarding policy mengizinkan.
- TCP reachability hanya precheck; MQTT success memerlukan broker evidence/CONNACK sukses, active subscription, publish/receive, auth/ACL, reconnect, dan log.

9.2 Data wajib untuk tiket IT

Data	Isi sekarang	Yang harus diisi sebelum submit
Wi-Fi STA MAC	Current source dapat publish MAC setelah online	Actual MAC per unit pada label/commissioning output sebelum onboarding
Broker/server endpoint	CHANGE_ME_MQTT_HOST	Production FQDN dan IP/reservation owner
Destination port	1884 bench	Final TCP port; TLS policy/certificate; protocol MQTT
Source network	Wi-Fi STA/DHCP inference	SSID, VLAN/subnet, DHCP/NAC/asset policy
DNS/NTP	TBD	Resolver/port and time source required for TLS/log correlation
Identity	Username/password provisionable	Per-device identity; rotation/revocation; topic ACL
Owner	TBD	LGU device PIC, Pertamina IT PIC, operations/HSE PIC, location/serial

9.3 Firewall/port matrix provisional

Source	Destination	Port	Purpose	Policy
Gateway VLAN	Broker FQDN/IP	TBD TCP (1884 bench)	MQTT uplink/downlink	Allow only registered Gateway; TLS/auth/ACL decision
Gateway VLAN	DNS	53 UDP/TCP if needed	Resolve broker	Site resolver only
Gateway VLAN	NTP	123 UDP if needed	Time/TLS/logs	Site NTP only
Operator/admin	Reverse proxy/dashboard	443 TCP proposal	Dashboard	HTTPS + RBAC; not from Gateway VLAN
Admin subnet	Node-RED admin	1880 TCP if retained	Restricted administration	Do not expose broadly; reverse proxy/VPN
Application server	MySQL dataset	3306 TCP if separate	Dataset recording	Local/internal only; never from Gateway VLAN
Any	Internet	None by default	No default requirement	Default-deny; exception via change approval

9.4 TLS gap dan pilihan keputusan

SECURITY-01 Gateway current source memakai WiFiClient plaintext, sedangkan generator server menolak broker non-loopback tanpa TLS dan credential. Karena itu claim 'TLS end-to-end ready' tidak benar untuk current source.

Opsi	Kelebihan	Konsekuensi	Rekomendasi
TLS end-to-end di Gateway	Auth/integrity/privacy on route; sejalan dengan server policy	Butuh CA trust, time sync, credential/cert per device, memory/performance, rotation/revocation test	Utama
Exception OT terisolasi	Dapat mempercepat pilot	Plaintext tetap ada; wajib VLAN/NAC/firewall/ACL/logging dan approval/risk acceptance	Sementara, formal only
Local bridge plaintext→TLS	Membatasi plaintext segment	Menambah asset, power, patching, monitoring, HA, dan another trust boundary	Transitional fallback

MAC allowlist/NAC membantu asset control, tetapi MAC dapat disalin dan tidak menggantikan authentication/authorization. Jika Pertamina mewajibkan WPA2-Enterprise/802.1X, current Gateway source belum membuktikan dukungan; perlu firmware change atau SSID/bridge exception yang disetujui.

9.5 Security acceptance

30. Approve architecture, asset inventory, MAC/FQDN/port/VLAN/DHCP/NAC/firewall/DNS/NTP/owner.
31. Test valid and invalid identity, unauthorized topic/command, certificate trust/expiry/revocation path, and audit log.
32. Verify dashboard via HTTPS/RBAC/session control; restrict Node-RED admin and DB.
33. Test Gateway/broker/Node-RED restart, queue/drop/reconnect, replay state, alarm burst, and no false ACK.
34. Perform 24–72 hour soak with evidence: timestamp, revision/hash, broker log, screenshot, negative tests, PIC, sign-off.
35. Use NIST OT security principles for segmentation, least privilege, logging, recovery, and change control; apply MQTT authentication/authorization/secure communication guidance.

10. Mounting, antena, kabel, grounding, dan site interface

10.1 Prinsip universal bracket

PROPOSAL MEKANIK Gunakan modular backplate terpisah untuk GLD1/CH, interchangeable clamp untuk round pole/handrail, wall/base plate untuk structure-approved anchor, serta bracket terpisah untuk solar panel dan antenna. Jangan freeze ukuran sebelum survey dimensional dan load approval.

- Candidate material: SS316L/passivated untuk coastal/chemical; gunakan galvanic isolation terhadap carbon steel/aluminium dan validasi corrosion/H₂S/salt-mist site.
- Slotted bracket + dual U-bolt/band clamp memberi adjustment untuk beberapa OD; wall/building memakai base plate/anchor yang disetujui structural owner.
- M8/M10/M12 dan spanner 13/17/19 mm hanya candidate standardization. Final bolt class, nut, washer, captive/locking method, torque, anti-seize, tool restriction, dan inspection interval harus disepakati.
- Tidak ada drilling, welding, hot-work, atau piggyback pada existing detector/support tanpa permit dan asset owner approval.
- Mechanical drawing harus memuat mass/CG, wind/seismic load, vibration, drain/orientation, service envelope, removal path, cable bend radius, gland location, ingress/drip loop, earthing point, and tag plate.
- GLD1 placement dekat existing sensor/power dapat mengurangi cable, tetapi tidak boleh mengganggu detector, calibration access, airflow, heat, or maintenance.

10.2 Tidak ada tinggi/distance universal

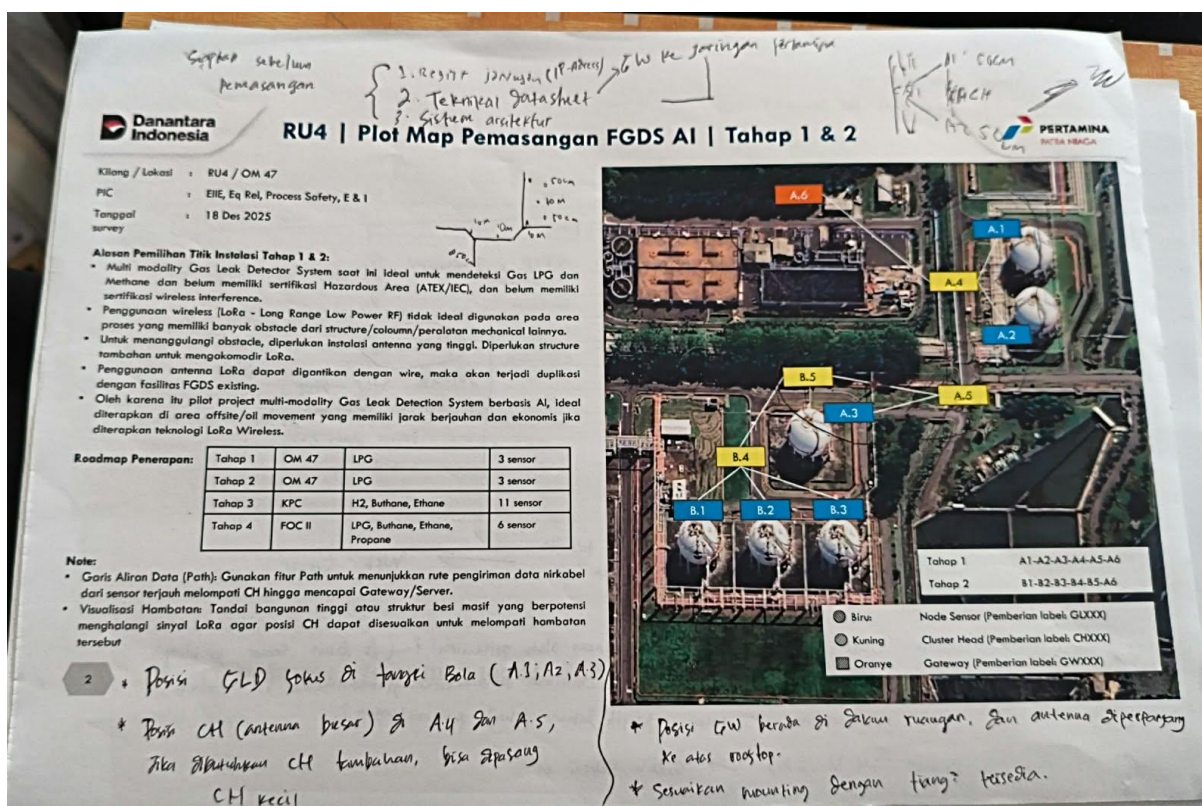
Ketinggian GLD1 tidak boleh ditentukan hanya dari nama gas. Densitas/buoyancy dipengaruhi temperature, pressure, release momentum, ventilation, obstruction, weather, and process geometry. Pertamina/HSE harus menyediakan hazard study/area drawing dan target release scenario; vendor kemudian memvalidasi coverage melalui dispersion/risk basis dan gas-release/chamber/site test. Catatan 'kebocoran hampir harian' bila benar harus diverifikasi dari incident/operations data—bukan alasan menempatkan prototype uncertified lebih dekat ke leak source.

10.3 Data survey mekanik wajib

Domain	Data minimum
Pole/handrail	OD range, material, wall thickness, height, existing load, allowable load, vibration, corrosion, clearance
Structure/wall	Anchor policy, concrete/steel condition, prohibited zones, drilling/welding/hot-work rules, structural sign-off
Environment	Ambient min/max, solar load, rain, flood, dust, salt, H ₂ S/chemical, wind/seismic, lightning class
Access	Work-at-height, ladder/scaffold/lifting, maintenance route, emergency egress, tool restriction, permit/JSA
Cable	Source panel, route/length, tray/conduit, cable type/area, gland/armor/shield, separation, fire rating, tagging

Domain	Data minimum
Antenna	Height, LOS/obstruction, separation, metal clearance, coax length/loss, rooftop access, grounding, lightning zone
Existing detector	Owner, no-interference distance, calibration envelope, mounting prohibition, power sharing approval

10.4 Visual evidence dan lokasi konsep



Gambar 4. Plot map/catatan lapangan. Ini input survey dan phase discussion, bukan approved plot plan, area-classification drawing, cable route, RF survey, atau installation permit.

11. Hazardous area, ATEX/IECEx, dan batas trial

SAFETY HOLD Tidak ditemukan certificate/marketing ATEX atau IECEx untuk assembly GLD1 maupun CH. Hingga konfigurasi final lengkap disertifikasi dan diterima Pertamina, status yang benar adalah: 'Engineering prototype — not certified for installation in a hazardous area.'

11.1 Apa arti ATEX/IECEx dalam konteks ini

- ATEX Directive 2014/34/EU mengatur equipment/protective systems/components untuk potentially explosive atmospheres di pasar EU; istilah ATEX bukan sinonim umum untuk 'explosion proof' dan belum otomatis menjadi satu-satunya acceptance basis Indonesia/site Pertamina.
- IECEx Certified Equipment Scheme memakai independent testing dan quality assessment. Ex component certificate masih memerlukan additional certification ketika diintegrasikan ke equipment/assembly.
- Area classification gas mengacu pada IEC 60079-10-1; selection/installation/documentation/initial inspection pada IEC 60079-14. Protection concept dapat melibatkan flameproof 'd', intrinsic safety 'i', atau lain-lain—harus dipilih oleh competent Ex design/certification route.
- SNI IEC 60079 editions dan ketentuan Migas/Pertamina yang diterima harus dikonfirmasi. Standard edition, certificate scheme, local approval, dan site procedure tidak boleh diasumsikan.
- IP/IK, material metal, physical partition, atau enclosure yang tampak industrial tidak sama dengan Ex certification.

11.2 Scope yang harus ikut sertifikat/assessment

Subassembly	Pertanyaan sertifikasi
GLD1 sensor head	Gas path, sensing element, hot surface, flame arrestor/flamepath, material, calibration access, contamination
GLD1 electronics/power	Protection concept, component ratings, spacing, thermal/T-class, input protection, fault analysis
CH electronics	Dual radio, charger, battery fault/thermal, enclosure, antenna feedthrough
Battery/solar	Cell/pack, protection, cable, panel, charger, connector, temperature, fault energy; whether inside certificate scope
Enclosure/gland/seal	Exact manufacturer/model, certificate schedule, special conditions, torque, cable range, blanking, ingress, flamepath
Antenna/surge/earth	RF output, antenna/cable/gland/surge/barrier configuration, bonding and installation conditions
Final assembly	Marking, gas group, T-class, EPL/category, ambient, certificate number, QAR/QAN, drawings/BOM revision
Installation	Area drawing, equipment selection, cable/earthing, initial inspection by competent person, dossier and maintenance

11.3 Pemisahan sensor dan power

Pemisahan electronics/power dari sensor dapat menjadi design direction, tetapi partisi biasa tidak cukup. Separation, seal/gland, flamepath, barrier, creepage/clearance, fault energy, thermal path, and every external connection harus berada dalam protection concept dan certificate/drawing scope.

Memindahkan trial dari LPG tank ke SRU juga tidak otomatis aman; SRU/H₂S tetap memerlukan area-classification drawing dan keputusan tertulis.

11.4 Hold points

36. Pertamina menerbitkan/menyetujui area-classification drawing untuk setiap titik: Zone/non-hazardous, gas group, T-class, EPL/category, gas, ambient, 3D boundary.
37. LGU/vendor memilih protection concept dan certification basis/edition yang diterima Pertamina/Migas/local authority.
38. Complete manufacturer BOM, controlled drawings, enclosure/gland/antenna/battery/solar schedule, risk/fault/thermal analysis selesai.
39. ExCB/laboratory application, quotation, test plan, schedule, certificate scope, QAR/QAN, and special conditions tersedia. Klaim '3 minggu' baru boleh digunakan setelah bukti ini.
40. Certificate authenticity/status diverifikasi pada controlled database; final marking cocok dengan as-built serial/revision.
41. Installation dossier, competency, permit/JSA, cable/earthing, inspection record, and maintenance plan disetujui.
42. Sebelum semua hold point classified ditutup, trial hanya pada lokasi non-hazardous yang ditetapkan Pertamina secara tertulis dan tetap mengikuti permit serta safety control.

12. Lokasi kandidat, fase, jadwal, dan pembagian kerja

12.1 Status lokasi

Lokasi/fase dari catatan	Gas/quantity	Status keputusan
OM47 tahap 1	LPG, 3 sensor	Printed plot input; handwritten A1/A2/A3 GLD1 dan A4/A5 CH; belum freeze
OM47 tahap 2	LPG, 3 sensor	Roadmap input; hazardous/site approval TBD
KPC tahap 3	H ₂ /butane/ethane, 11	Roadmap input; scope/contract TBD
FOC II tahap 4	LPG/butane/ethane/propane, 6	Roadmap input; scope/contract TBD
SRU/PSU candidate	H ₂ S/NH ₃ mentioned	Relocation/trial discussion; not automatically safe; classification TBD
Sampit candidate	LPG/methane/H ₂ S mentioned	Meeting note only; site/scope/date TBD

12.2 Rencana kerja berbasis tanggal

Tanggal target	Deliverable	Owner utama	Exit criteria
20 Aug 2026	Internal decision pack	LGU	Enam keputusan disetujui; P0/P1 owner ditetapkan
21–25 Aug	RFI terkonsolidasi	LGU → Pertamina/vendor	Network/HSE/power/mechanical/server/contract questions terkirim
≤28 Aug	Datasheet Rev A + measurement plan	GLD1/CH/GW/server leads	Controlled fields, TBD register, FAT instruments/samples
1–11 Sep	Pre-survey pack	LGU + Pertamina	Point list, drawing request, permit/JSA, survey template, IT ticket draft
Next site visit	Dimensional/HSE/power/RF/network survey	Joint team	Signed measurements, photos, routes, classification, RF/network evidence
≤18 Sep	Freeze phase-1 interface	LGU + Pertamina	Location, pole, cable, power, antenna, network, owner approved
≤25 Sep	Support/cable/pole submission	LGU mechanical/electrical	Drawing/BoQ before end-Sep target; assumptions closed
Oct 2026 onward	Design closure + certification + FAT	LGU/vendor/ExCB	P0 closed; certification/test milestones baselined
Before Sep 2027	SAT readiness for 3 GLD1 + CH	Joint program	Contract/scope, certificate, network, installation, training, spares, FAT pass
Sep 2027 target	Phase-1 field acceptance	Pertamina signatory	SAT evidence + punch-list closure; target must be confirmed

12.3 Pembagian tanggung jawab

Deliverable	LGU	Pertamina RU IV / IT / HSE	Vendor/ExCB
Controlled device datasheets	R/A: content, BOM, revision, evidence status	C: site requirement/approval	C: vendor/certificate data
Area classification & acceptance basis	C: design input	R/A: site drawing, gas/T-class/EPL, procedure	C: certification route
GLD1 power design/FAT	R/A: protection, sizing, fan/alarm, test	C: supply/cable/permit	C: PSU/part data
CH energy/mechanical	R/A: panel/cell/profile/bracket	C: autonomy, pole/load, environment	C:

Deliverable	LGU	Pertamina RU IV / IT / HSE	Vendor/ExCB
			battery/solar/enclosure certificates
Gateway/network	R: MAC/device/config/test support	A: VLAN/NAC/firewall/FQDN/port/TLS policy	C: antenna/surge
Server deployment	R: package/flow/test/runbook	A/C: platform/security/operations	C: platform support
Certification	R: controlled design dossier and samples	A/C: accepted scheme/site interface	R/A: assessment/test/certificate in own scope
Installation/SAT	R: commissioning/test evidence	A: permit, execution owner, sign-off	C: special conditions/inspection support

R = Responsible, A = Accountable, C = Consulted. Nama orang, kontrak boundary, procurement owner, and signatory masih TBD dan harus menggantikan role generik sebelum distribusi eksternal.

13. RFI/pernyataan yang harus ditutup

13.1 Kepada Pertamina HSE/site

43. Berikan area-classification drawing untuk A1–A5, OM47, SRU/PSU, dan candidate lain: Zone/non-hazardous, gas/dust group, temperature class, EPL/category, actual gas, ambient, dan 3D boundary.
44. Tentukan scheme/standard edition yang diterima: IECEx/ATEX/SNI IEC 60079/Migas/internal; minimum IP/IK, corrosion/H₂S/salt-mist, EMC, and signatory.
45. Apakah battery, solar panel, gland, antenna, surge protector, cable, and complete assembly wajib dalam certificate scope?
46. Konfirmasi incident/leak basis, target detection scenario, existing detector interface, release/dispersion information, and permitted non-hazardous trial location.
47. Tetapkan permit/JSA, work-at-height, hot-work/drilling/welding restriction, gas testing, competent Ex personnel, initial inspection, dan maintenance dossier.

13.2 Kepada Pertamina electrical/mechanical

48. Konfirmasi apakah source '110VA' sebenarnya 110 VAC; berikan voltage/frequency/phase/earthing, panel, breaker, spare capacity, UPS/emergency status, dan cable route.
49. Berikan pole/handrail/building OD/material/thickness/height/allowable load/wind/seismic/vibration/corrosion serta anchor policy.
50. Setujui allowable bolt/nut/washer/material/torque/anti-loosening/tool size dan inspection interval.
51. Berikan cable standard, conductor/armor/shield/gland/termination/tagging/fire rating/segregation, allowable drop, and grounding/bonding/lightning rules.
52. Konfirmasi antenna rooftop access, height/clearance, lightning zone, coax route/max length, grounding, work-at-height, and structural approval.
53. Tetapkan target autonomy CH, availability, maintenance access, panel cleaning interval, shading/solar data, dan allowed downtime.

13.3 Kepada Pertamina IT

54. SSID/band dan authentication: PSK atau 802.1X; captive portal dilarang/ditangani bagaimana?
55. VLAN/subnet, DHCP reservation/static policy, gateway, DNS, NTP, NAC/MAC allowlist, asset registration, and location/owner fields.
56. Broker FQDN/IP, final TCP port, TLS/mTLS/VPN policy, CA/certificate issuance/rotation/revocation, per-device identity, and topic ACL.
57. Firewall matrix, internet default-deny, logging/SIEM/retention, connection monitoring, incident response, and change window.
58. Approved server platform: OS/version, VM/physical, CPU/RAM/storage, network zone, backup, monitoring, patching, RPO/RTO/HA.

- 59. Dashboard HTTPS/RBAC/SSO/session policy; admin subnet; Node-RED and DB exposure restrictions.
- 60. Acceptance evidence format: CONNACK/broker logs, negative tests, packet capture authorization, screenshot, timestamp/NTP, PIC, and sign-off.

13.4 Kepada vendor/desain

- 61. Complete manufacturer BOM untuk GLD1/CH/GW: exact part/model/lot termasuk MQ, fan/alarm, PSU, battery, panel, enclosure, glands, antenna/coax/surge/bracket.
- 62. Gerber, fabrication/assembly drawing dalam mm, hole table, STEP/3D, mass/CG, enclosure drawing, gasket/gland/torque/cable schedule.
- 63. GLD1: selesaikan F1 untuk input 24 V, MQ-7 heater cycle, input transient/TVS/thermal, fan/alarm current, dan identitas sensor per socket.
- 64. CH: exact cell/pack and panel data; compatibility BQ25185; 24 h current profile; autonomy/recharge/brownout/dual-radio/thermal test; IEC 62133-2/UN 38.3 evidence.
- 65. RF: antenna/coax/gland/surge datasheet, gain/EIRP calculation, dual-radio isolation, link budget, and site acceptance criteria RSSI/SNR/package-loss.
- 66. If Ex is claimed: CoC/DoC/QAR/QAN, certificate schedule, special conditions, marking, ambient, gas group, T-class, EPL, drawing/BOM revision, and online verification record.

13.5 Scope/contract questions

- 67. Apakah phase 1 benar 3 GLD1 + 1 CH, dan apakah Gateway/server termasuk supply/installation/operation scope?
- 68. Apakah additional contract diperlukan untuk site survey, pole/cable/support, certification, server/network, FAT/SAT, training, spares, and maintenance?
- 69. Siapa accountable owner dan signatory untuk HSE, IT, electrical, mechanical, operations, procurement, cybersecurity, certification, FAT, and SAT?
- 70. Konfirmasi target end-September 2026 dan September 2027 beserta dependency, review duration, approval lead time, and change control.
- 71. Tentukan definition of done, warranty/support, spare strategy, calibration interval, data ownership/retention, and post-SAT operational handover.

14. FAT, SAT, dan bukti penerimaan

14.1 FAT matrix

Domain	Test minimum	Pass evidence
GLD1 power	20 cold-start; 120 min profile; all heaters; LoRa/RS485/LED/alarm/fan; cable drop; +5 V/3,3 V; closed-enclosure thermal	Waveform/log calibrated instrument; rail within MPN limit; no reset/current-limit; ≥20% headroom; blocker closed
GLD1 sensing	MPN/lot; heater resistance/cycle; channel map; zero/span; target/interferent; humidity/temp; exposure/recovery; reference instrument	Calibration/dataset report; confusion/false alarm/miss criteria; traceability
CH energy	24 h production/test cadence; idle/RX/TX/dual peak; charge/day/night; autonomy; recharge; brownout; temp	Energy budget; measured profile; cell/panel BOM; autonomy/recovery pass
Mechanical	Fit/retention; fastener torque; vibration; pull/load; ingress; corrosion basis; cable/gland; 2 RF feedthrough	Drawing/revision, photos, torque sheet, inspection/test report
RF	Bench protocol; range topology; interference; packet/sequence/retry; dual-radio isolation	RSSI/SNR/PER/latency capture; no unaccounted loss
GW/server	Provisioning, auth/TLS/ACL, normal/alarm/downlink, disconnect/restart/replay, 24–72 h soak, dashboard/security/backup-restore	Broker/server logs, flow hash, screenshot, negative test, restore proof
Hazardous	Certification test/assessment per agreed scheme	Valid certificate/QAR/markings/drawing/BOM special conditions; not self-declared

14.2 SAT matrix

Stage	Checks	Evidence/sign-off
Pre-install	Approved drawing/BOM/certificate; serial/markings; permits/JSA; classification; cable/pole/power/network readiness	Release note and hold-point signatures
Mechanical/electrical	Mount/orientation/torque/load/access; gland/drip/ground/bond/surge; 24 V polarity/no-load/full-load/drop; CH panel/battery	As-built photos/drawing, torque/power sheet
RF/network	Actual height/location; RSSI/SNR/PER; Gateway IP/DNS/NTP; TCP precheck; MQTT CONNACK/subscriptions/auth/ACL	RF capture, broker log, IT sign-off
End-to-end	GLD1→CH→GW→broker→Node-RED→dashboard; alarm; downlink/result; timestamp/sequence; disconnect/recovery	Correlated event pack; no false ACK
Safety	Certificate/config match; special conditions; initial Ex inspection; surface temperature/ground; no interference existing detector	Competent inspector record/HSE sign-off
Soak/handover	24–72 h; stale/drop/reconnect; backup/restore; operator training; spares/calibration/maintenance	Acceptance report, punch list, owners/dates

14.3 Evidence package per test

- Test ID, requirement/source, device serial, HW/FW/flow/config revision dan hash.
- Date/time/timezone, synchronized clock source, location/zone, environmental condition.
- Instrument maker/model/serial/calibration status; setup drawing and cable/antenna/power details.
- Raw log/capture, calculation, screenshot/photo, expected/actual, pass/fail, anomaly, retest reference.
- PIC executor/reviewer, Pertamina witness/signatory, open punch item, owner, due date.

- Explicit separation: static-source evidence, build evidence, hardware measurement, network/MQTT evidence, deployment revision, and field acceptance.

14.4 Definition of readiness

Gate	Ready bila
Ready for internal design review	This report approved; owners/dates assigned; source/BOM/drawings controlled; P0 plan funded
Ready for non-hazardous pilot	Pertamina written non-hazardous point; permits; FAT power/RF/network/server pass; mechanical/electrical survey closed
Ready for classified-area installation	Area classification + accepted complete-assembly certification + installation design + competent initial inspection + HSE/IT/site approvals
Ready for production handover	SAT/soak/backup/restore/security/training/spares/calibration/maintenance/ownership pass; punch list controlled

15. Daftar sumber dan register bukti

15.1 Sumber lokal primer

ID	Sumber	Penggunaan dan batas
L-01	C:\Users\MSI\Downloads\Summary recorder 081426 .pdf	Notulen/meeting input; bukan sertifikat atau acceptance
L-02	C:\Users\MSI\Downloads\cilacap\ (7 gambar unik)	Konsep enclosure, plot map, catatan kebutuhan; bukan dimensional/Ex/RF proof
L-03	C:\Users\MSI\Downloads\CH.zip	Nested EasyEDA schematic/PCB inventory; static design only
L-04	docs\wiring\gld-project-ver2-2026-07-01\source-GLD_Project.zip	GLD1 full schematic/PCB static
L-05	firmware\config*.h; firmware\gld\ch\gateway\src; platformio.ini	Current source configuration/behavior; not build/flash/runtime proof
L-06	server\nodered\ generator\snapshot\functions\README	Server source and read-only drift result; not live deployment proof
L-07	docs>manual and output\pdf old architecture/task documents	Cross-check only; new report restates all needed content

15.2 Current-source evidence pointers

Area	Path/line pointer
GLD1 power/profile	firmware\gld\src\GldUnifiedMain.cpp:5428; source-GLD_Project.zip full EasyEDA schematic/PCB
CH radio/power/runtime	LoraStarConfig.h:13–49; LoraMeshConfig.h:13–48; ChConfig.h:35–40,77–92; ChStarMeshRuntimeMain.cpp:360–383,1054–1065
CH active profile	firmware\platformio.ini:287–309,332–337; ChBoardPinsCh3.h:26–31
GW Wi-Fi/MQTT	GatewayMqttMeshMain.cpp:238–241,636–655,797–870,1589–1686,1759–1763
GW config	ServerConfig.h:36–64; GwConfig.h:19–35; FirmwareVersion.h:5–14; platformio.ini:363–380
Server security/drift	apply-pertamina-gld-flow.js:105–116,2212–2224,2272–2288; pertamina-gld-decode.js:1023–1130
Server capability	server\nodered\README.md:1–98; apply-pertamina-gld-dataset-flow.ps1:1–16,32–53

15.3 Sumber primer vendor/standard

R-01 ATEX Directive 2014/34/EU: <https://eur-lex.europa.eu/eli/dir/2014/34>

R-02 Workplace explosive atmospheres Directive 1999/92/EC: <https://eur-lex.europa.eu/eli/dir/1999/92/oj>

R-03 IECEx Certified Equipment Scheme overview: <https://www.iecex.com/certified-equipment-scheme/overview/>

R-04 IECEx FAQ / certificate verification guidance: <https://www.iecex.com/resources-and-news/frequently-asked-questions/>

R-05 IEC 60079-10-1:2020 area classification: <https://webstore.iec.ch/en/publication/63327>

R-06 IEC 60079-14:2024 design, selection, installation, initial inspection: <https://webstore.iec.ch/en/publication/66049>

R-07 IEC 60079-0:2026 general requirements: <https://webstore.iec.ch/en/publication/71519>

R-08 IEC 60079-1 flameproof enclosure d: <https://webstore.iec.ch/en/publication/621>

R-09 IEC 60079-11:2023 intrinsic safety: <https://webstore.iec.ch/en/publication/60654>

R-10 IEC 60079-17 inspection/maintenance: <https://webstore.iec.ch/en/publication/631>

R-11 BSN SNI IEC 60079 catalog: <https://pesta.bsn.go.id/produk/index/471>

R-12 NIST SP 800-82 Rev.3 OT Security: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

R-13 OASIS MQTT Version 5.0: <https://docs.oasis-open.org/mqtt/mqtt/v5.0/mqtt-v5.0.html>

R-14 TI BQ25185 datasheet: <https://www.ti.com/lit/ds/symlink/bq25185.pdf>

R-15 TI TPS63020: <https://www.ti.com/product/TPS63020>

R-16 TI LMR51450: <https://www.ti.com/product/LMR51450>

R-17 Littelfuse miniSMDC resettable PTC datasheet:

https://m.littelfuse.com/~media/electronics/datasheets/resettable_ptcs/littelfuse_ptc_minismdc_datasheet.pdf.pdf

R-18 Espressif ESP32-S3-WROOM-1/1U datasheet: https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

R-19 EBYTE E22-900MM22S: <https://www.cdebyte.com/products/E22-900MM22S>

R-20 IEC 62133-2 battery safety: <https://webstore.iec.ch/en/publication/32662>

R-21 UN 38.3 test summary resources: <https://unece.org/transport/dangerous-goods/rev8-files>

15.4 MQ reference datasheets

Winsen MQ-2: <https://www.winsen-sensor.com/d/files/manual/mq-2.pdf?v=1.0.24>

Winsen MQ-3B: <https://www.winsen-sensor.com/d/files/manual/mq-3b.pdf?searchid=6821>

Winsen MQ-4: <https://www.winsen-sensor.com/d/files/semiconductor/mq-4.pdf>

Winsen MQ-5: <https://www.winsen-sensor.com/d/files/manual/mq-5.pdf>

Winsen MQ-6: <https://www.winsen-sensor.com/d/files/manual/mq-6.pdf>

Winsen MQ-7B: <https://www.winsen-sensor.com/d/files/manual/mq-7b.pdf>

Winsen MQ-8: <https://www.winsen-sensor.com/d/files/semiconductor/mq-8.pdf>

Winsen MQ135: https://www.winsen-sensor.com/d/files/mq135-%28ver1_6%29---manual.pdf

Semua MQ source di atas hanya reference candidate. Manufacturer/MPN/variant/lot yang benar-benar terpasang belum ditemukan pada EasyEDA/BOM/repo.

15.5 Pernyataan akhir audit

TRACEABILITY Setiap klaim final harus menunjuk salah satu: controlled source, controlled drawing/BOM, build artifact, measurement record, network/broker log, deployment revision, certificate database, atau signed FAT/SAT. Ketika evidence itu belum ada, status tetap TBD/NO-PROOF—bukan diasumsikan lulus.

— Akhir laporan —